AML Project · Phase 1 01

Neuro-Symbolic Causal Inference

Teaching a model to tell correlation from causation.

TEAM

Krishnadevan Manju Liby · Hemanth Tenneti · Arina Ali · Manan Bansal · Aaloke Eppalapalli

Newton School of Technology · Rishihood University

Problem + data 02

LLMs confuse correlation with causation

Corr2Cause (Jin et al., ICLR 2024) tests exactly this: given correlations, must a causal claim be true? Even fine-tuned transformers mostly pattern-match.

EXAMPLE PUZZLE

A moves with C

B moves with C

A and B are unrelated

Claim: “A causes C” TRUE

only one graph fits: A → C ← B

205,734
training examples

21%
of claims are valid

0
premises shared between train and test

6,943
buggy labels corrected (3.3%)

Method 03

Three steps: read, reason, answer

01

Read

Regex rules turn each sentence into a fact: “A correlates with C” → correlates(A, C).

language layer

02

Reason

The PC algorithm finds every causal graph that fits the facts (the Markov equivalence class). Support = share of those graphs where the claim holds.

PC · Meek rules · support

03

Answer

Valid claims form a DAG → Bayesian network (pgmpy) → answers do(X) “what if?” questions.

do-calculus

23 features, one readable rule

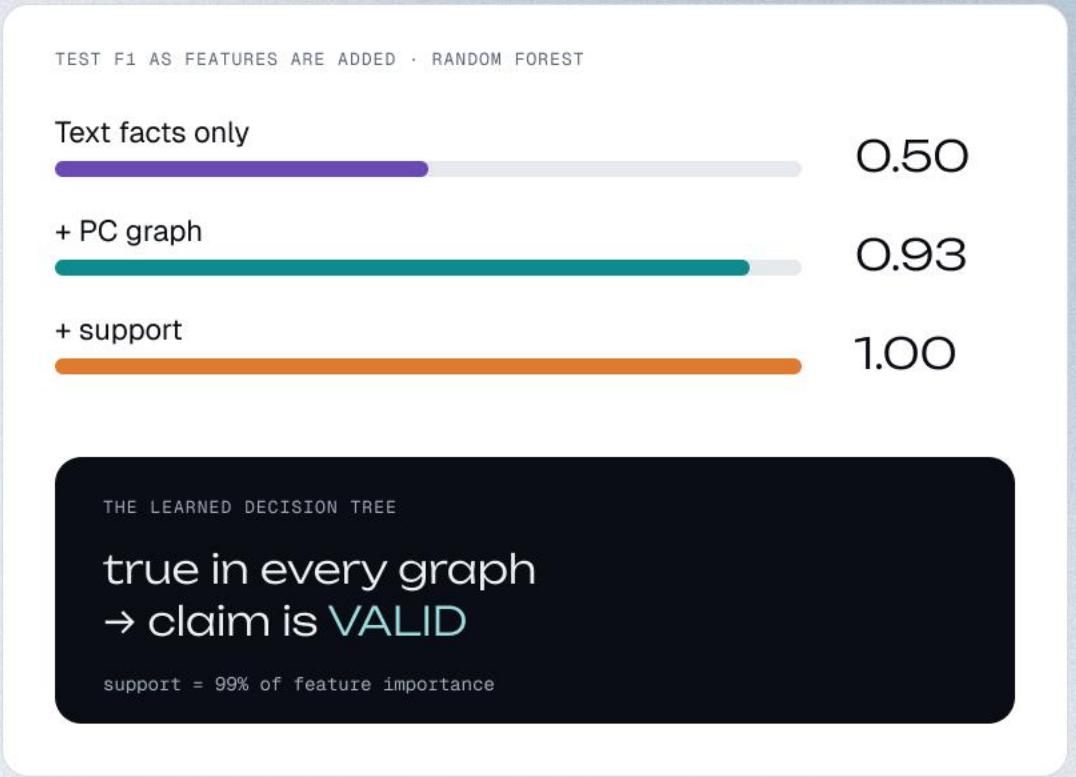
4 **Relation type**
direct · indirect · collider · confounder

8 **Text facts**
what the premise says about X and Y

9 **PC graph**
edges, ancestors, common parents

2 **Support**
share of equivalent graphs where the claim holds

Models compared: Logistic Regression · Decision Tree (depth 4) · Random Forest (300 trees)



Near-perfect on puzzles, lost on reworded ones

F1 · VALID CLASS	NORMAL (TEST)	RENAMED LETTERS	REWORDED
Text-only baseline (TF-IDF + LR)	0.655	0.322	0.250
Our pipeline (Rules + PC + Tree)	0.997	0.997	0.000

73 / 73

causal links recovered, no false edges

97.5%

of 242 “what if?” questions answered correctly

0.73 vs 0.49

seeing vs doing: hidden causes inflate the observed effect

Next: Phase 2 06

Fix the weak spot with DistilBERT

Regex reads 0% of reworded claims, so the reasoning never runs. Phase 2 keeps the reasoning and swaps only the reader.

2A · End-to-end

DistilBERT reads the puzzle and answers directly. Our deep-learning baseline.

2B · Hybrid (ML + DL)

DistilBERT reads, PC + tree reason. Justified by the 0% gap; same evaluation code.

Validation: early stopping on dev F1 · Regularization: weight decay · Ablation: ± variable-renaming augmentation

Thank you 07

Questions?

Correlation is cheap.
Causation needs reasoning.

Krishnadevan Manju Liby
Lead · graph stage & Bayesian network

Hemanth Tenneti
Data, labels & EDA

Arina Ali
Features, models & ablation

Manan Bansal
PC algorithm & theory

Aaloke Eppalapalli
Language layer & robustness sets

github.com/HemanthTenneti/AML-DL-Project-2026

Where this work fits

Causal graphs — Pearl (1988, 2009)

DAGs, d-separation and the do-operator turn “what if?” into probability.

Causal discovery — Spirtes et al.; Meek; Verma & Pearl

The PC algorithm recovers structure from independence facts, up to Markov equivalence.

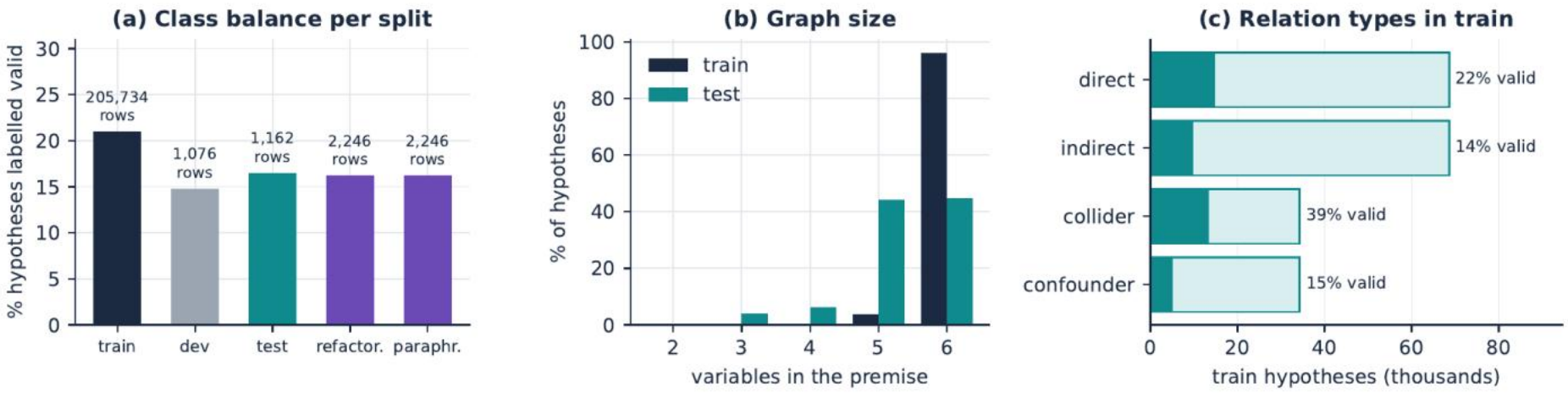
Corr2Cause — Jin et al., ICLR 2024

LLMs and fine-tuned transformers fail to infer causation from correlation and break under rewording.

Shortcut learning — Gururangan 2018; Geirhos 2020

Text models exploit dataset artefacts, matching our TF-IDF collapse under renaming.

Data quality and EDA



Premise-level split

0 premises shared between train and test, so no near-duplicate leakage.

6,943 labels fixed (3.3%)

A generator bug in the original data; we use the published corrected labels.

0.949 on original labels

Test F1 with the buggy labels, reported for transparency.

The theory in three ideas

Markov equivalence

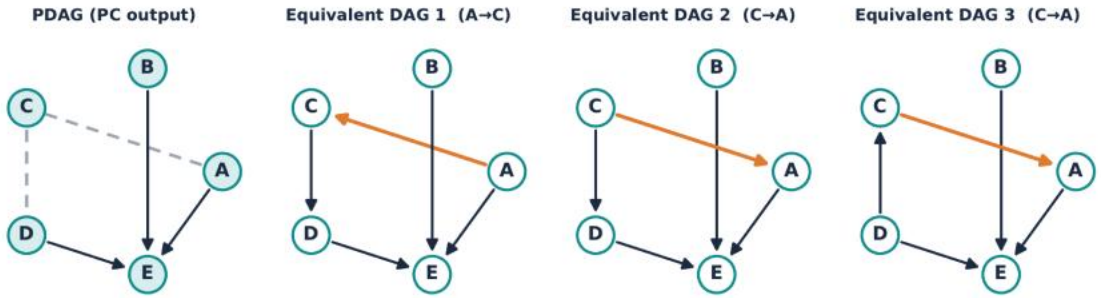
Independence facts fix the skeleton and colliders, not every arrow. Several DAGs can fit the same facts.

Support criterion

A claim is entailed only if it holds in every equivalent DAG (support = 1).

do-calculus

P(Y | do(X)) cuts X's incoming edges, so confounders stop leaking into the effect.



All 23 features

Relation · 4

- rel_direct
- rel_indirect
- rel_collider
- rel_confounder

Text facts · 8

- n_vars
- marginally_correlated
- adjacent
- n_sepsets
- min_sepset_size
- degree_x
- degree_y
- n_common_neighbours

PC graph · 9

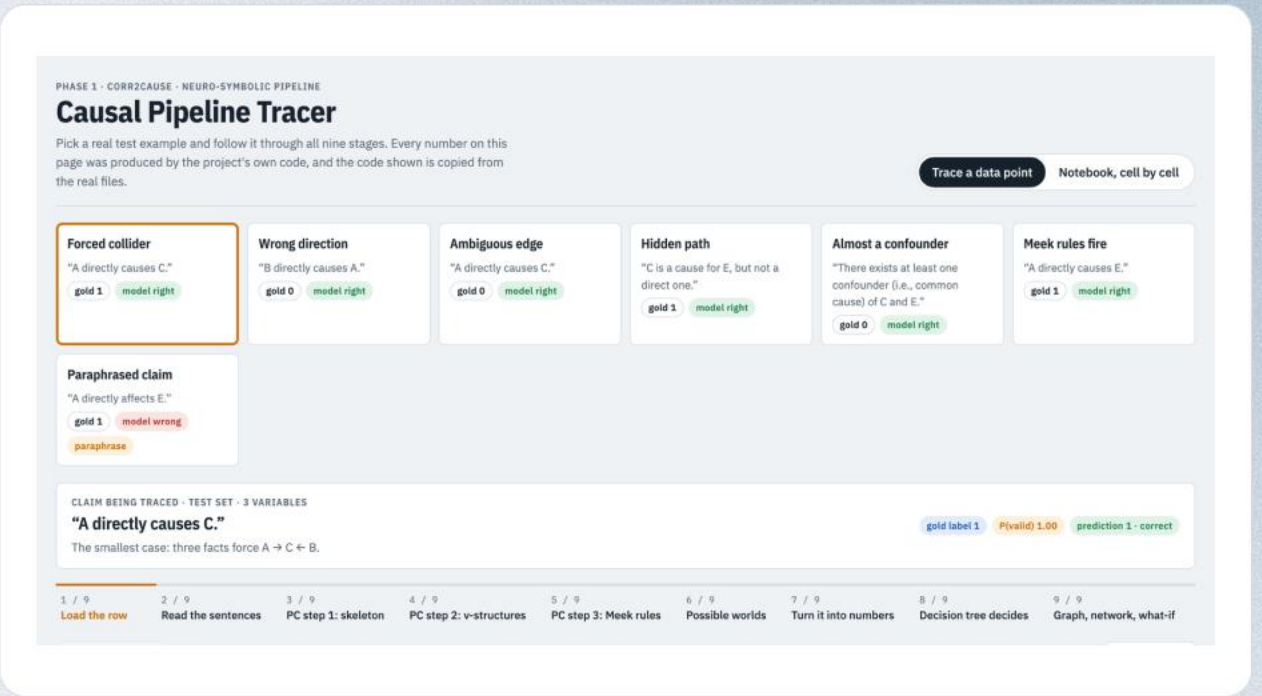
- edge_x_to_y
- edge_y_to_x
- edge_undirected
- x_ancestor_of_y
- y_ancestor_of_x
- x_possible_ancestor_of_y
- y_possible_ancestor_of_x
- n_common_children
- n_common_parents

Support · 2

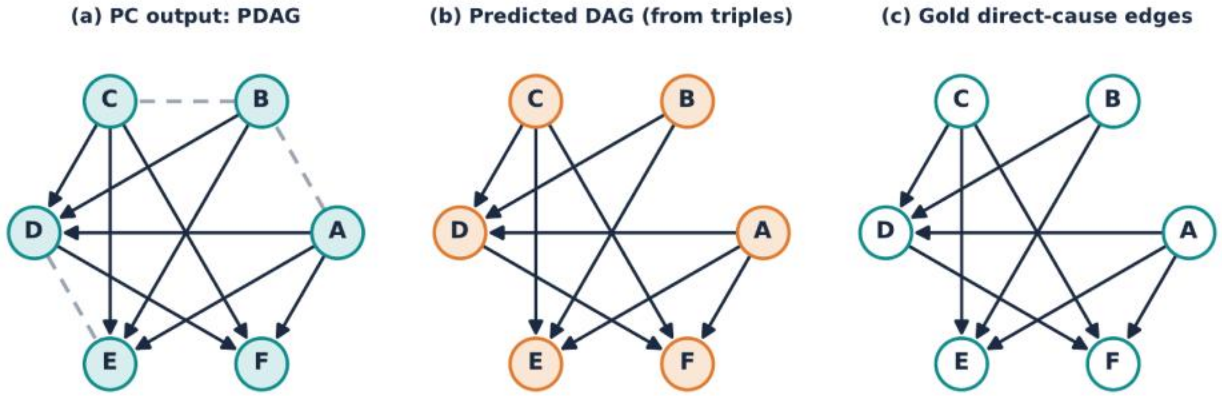
- support
- n_equivalent_dags

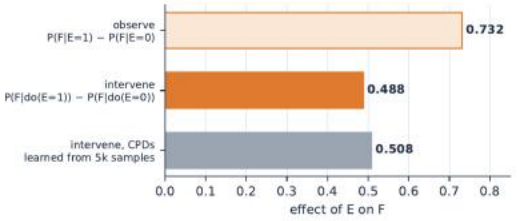
Reproducible and tested

- **14 unit tests**
  `pytest: parsing, PC rules, support, graph stage`
- **One notebook**
  `regenerates every number and figure`
- **Pipeline tracer**
  `docs/pipeline_visualiser.html`
- **Repository**
  `github.com/HemanthTenneti/AML-DL-Project-2026`



Graph stage evidence





Observed effect 0.732 vs intervened 0.488. With CPDs learned from 5k samples, pgmpy recovers 0.508.

6 query misses: chains through edges whose direction correlations cannot identify.